

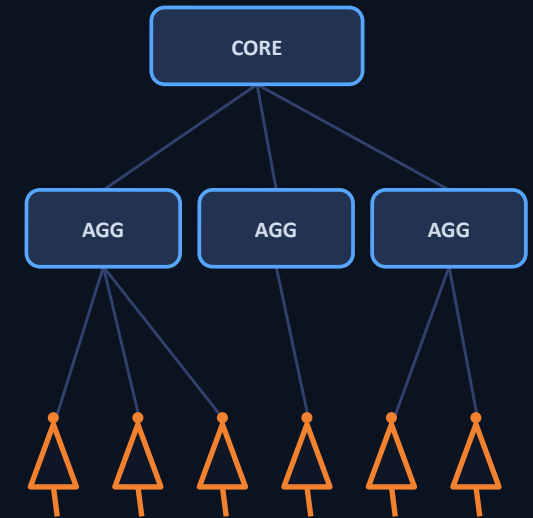
RAN & Transport Isolation Module

Rank the sites, nodes and transport paths most impacted by transport-network problems — and tell you which are transport vs RAN, with the evidence.

IP Transport Analytics

Impact Analytics Dashboard

Phase-2 Audit Prioritisation



Synthetic-data build · standalone web app · github.com/manjunathsept11/ran-transport-isolation

Correlate experience with transport — then rank and attribute

Ingest and correlate end-user experience metrics with transport-layer telemetry to identify and rank the sites, nodes and transport paths most impacted — and separate transport-attributed impairment from RAN-attributed impairment using serving-cell context and radio signal quality.

5

data feeds
stitched

7

analytics
modules

75–100

sites for the
Phase-2 audit

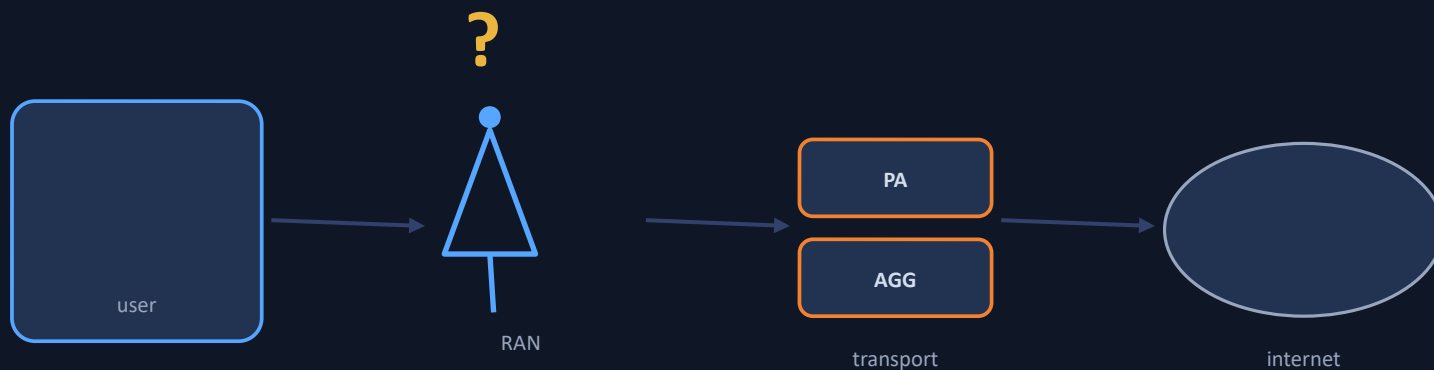
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LLM calls in
the runtime

Deliverables

- Impact analytics dashboard ranking sites, nodes and transport paths
- Headline KPIs: TCP RTT, TCP Fail %, Ookla throughput, VoNR MOS, YouTube QoE
- Prioritised list of the top 75–100 sites for a Phase-2 field audit (CSV / Excel)
- Per-site root-cause indicators and a recommended technician action
- Transport-attributed impairment distinguished from RAN-attributed

Users complain. Whose problem is it?



Looks like RAN

- Coverage hole / poor RSRP
- Interference, low RSRQ
- PRB exhaustion at busy hour
- Sleeping sector, VSWR / feeder fault

Looks like Transport

- Fibre degradation, microwave fade
- Backhaul congestion at busy hour
- SFP / CRC errors, queue drops
- Routing flap, MTU black-hole

Same symptom — high latency, buffering, dropped calls — two different owners, two different fixes.

Manual, siloed, and unmeasurable

● Siloed data

Ookla, N3 probes, TWAMP, SevOne each live in their own tool — no single correlated view.

● Manual triage

Engineers eyeball KPI graphs site by site. It does not scale to a whole market.

● Finger-pointing

RAN and transport teams each say 'not us'; tickets bounce between them for days.

● No prioritisation

Which 80 sites do we audit first? Today that answer is gut feel.

● No ground truth

There is no way to check whether the triage call was actually right.

● Slow Phase-2 audits

Shortlisting sites for field work takes weeks of cross-team analysis.

Five feeds, one timeline



Ookla SpeedTest

user experience + radio

UL/DL throughput, loaded latency, RSRP, RSRQ



N3 probe EDRs

session — the split point

TCP client RTT, TCP server RTT, TCP fail %, retransmissions, VoNR MOS



YouTube / Audio QoE

application

video / audio MOS, rebuffer ratio, startup delay



TWAMP

pure transport (active probes on each link)

one-way / round-trip delay, frame loss, jitter



SevOne

transport devices

router queue depth, discards, interface utilisation, CRC errors

Joined on site · cell · transport-path · 5-minute bin.

From five raw feeds to one feature table



One pipeline — batch now, streaming later



Now — batch

- Synthetic generator writes the SQLite warehouse + Parquet raw layer
- na generate → na analytics → na report (or the Settings page)
- Dashboard reads pre-computed serving tables — always fast

Live — streaming (Phase 2)

- The five feeds land continuously via the data-acquisition / ingestion layer
- 5-min rollups on arrival; analytics recompute on a schedule (e.g. hourly)
- Same serving tables, same dashboard — plus alerting on new priority sites

SQLite warehouse · FastAPI backend · React dashboard · background job runner

Transport vs RAN — the tell

1 The RTT split

TCP client RTT = radio + the whole transport path (round trip). TCP server RTT = internet only, site-independent — the control.

Client RTT up while server RTT flat → access / transport, not the internet.

2 Signature match

Radio: RSRP / RSRQ degraded, PRB near 100 %, usually one sector. Transport: TWAMP loss / jitter, SevOne queue / CRC, per-link delay — the transport layer's own instruments.

Pure-transport instruments lit, radio clean → transport.

3 Common cause

A single radio fault cannot degrade 18 sites at once. A shared backhaul link does it every time.

'% of sibling sites on the shared uplink also degraded' is the strongest tell.

transport

RAN

shared

unclear

+ a calibrated confidence score

Any one alone is weak — together they give a class and a confidence.

Impact scoring — which sites, in what order

Per-KPI severity

robust exceedance of each headline KPI vs its baseline, session-weighted

Composite score

normalise each severity, then a weighted blend over the 5 headline KPIs

Impact weighting

× sessions impacted × users impacted (from the raw session records)

Worst window

the 6-hour span with the largest summed degradation — where to look

Priority slice

top ≈ 75–120 by impact → flagged for the Phase-2 audit, exportable to Excel

Ranking is deliberately transparent — a weighted composite, not a black box — so an engineer can see why a site is at #3.

Rules ∪ LightGBM + SHAP, calibrated ensemble

Rule engine

deterministic thresholds on the robust-z features; separates pure-transport signals from ambiguous ones; emits a class + a plain-English evidence list

LightGBM classifier

gradient-boosted trees, multiclass, class-weight balanced, trained on the injected ground-truth labels per (site, hour)

SHAP

Shapley values on the booster → the top features that drove each per-site call, shown as bars

Calibrated ensemble

isotonic-calibrated; rule + ML combined, with a guardrail so a confident RAN rule beats an over-eager transport ML vote

Evaluated against ground truth every run: precision / recall / F1 per class, confusion matrix — reported in the dashboard and the notebook.

Anomaly · correlation · variability

Anomaly detection

- STL seasonal-residual + robust threshold, per KPI series
- Isolation Forest + PCA reconstruction error, multivariate
- clustered by time + topology into candidate incidents
- matched to ground truth by site-set IoU

Correlation

- Spearman + partial correlation between KPI pairs
- lead / lag cross-correlation — finds precursors
- e.g. SevOne queue buildup leads the QoE drop
- per-site layer driver split (transport vs radio)

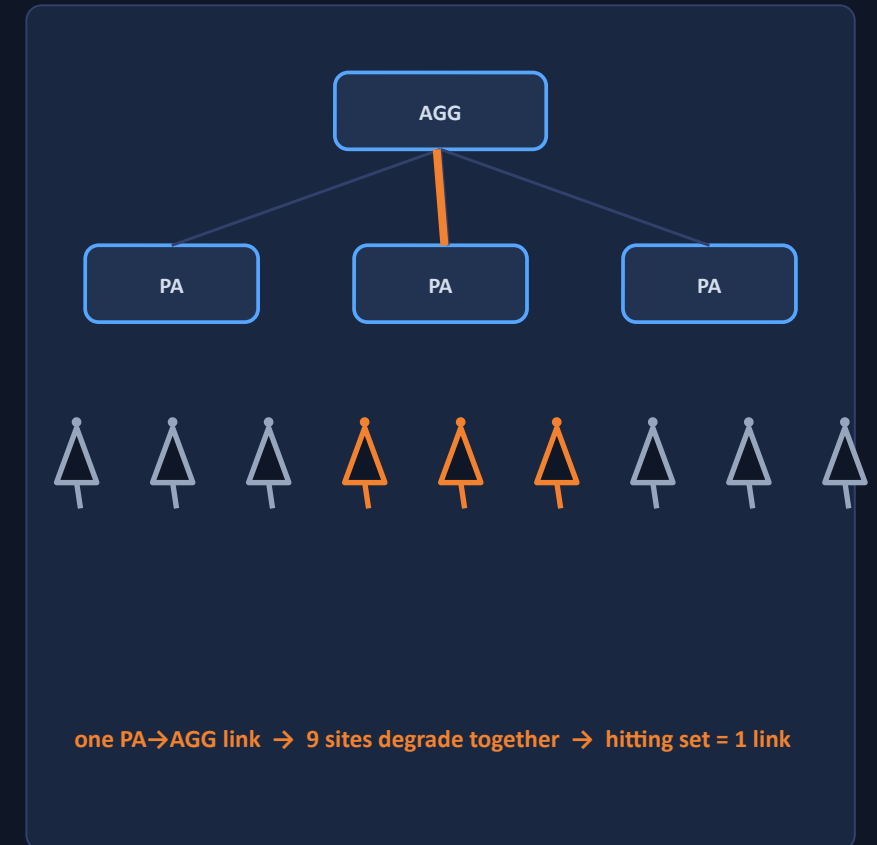
Variability

- variance components: site / hour / day / residual
- CV, IQR, busy-hour vs off-peak spread
- PSI drift, week over week
- flags high-variance / acceptable-mean sites

Localise the fault to a link

The transport network is a tree in a networkx graph. Given the set of sites degraded together in one window, a greedy minimal hitting set finds the smallest set of shared links whose failure explains every one of them — a Noisy-OR-style fault localisation.

- Combines attribution + anomaly cluster + changepoint timing + topology
- Confidence = fraction of affected sites the hitting set explains
- Matched to ground truth (demo) and given a recommended technician action



The data is synthetic — on purpose

Why

- Data-pipeline engineering is not in scope yet — real feeds arrive in Phase 2.
- A simulator lets us build and validate the analytics against known answers.
- Labelled fault injection → every record has a ground-truth cause.
- So precision / recall of the attribution is a real, reported number.

The layered causal model

Demand

diurnal + weekly load shape

Radio

path loss → RSRP · load → PRB · interference → RSRQ

Transport

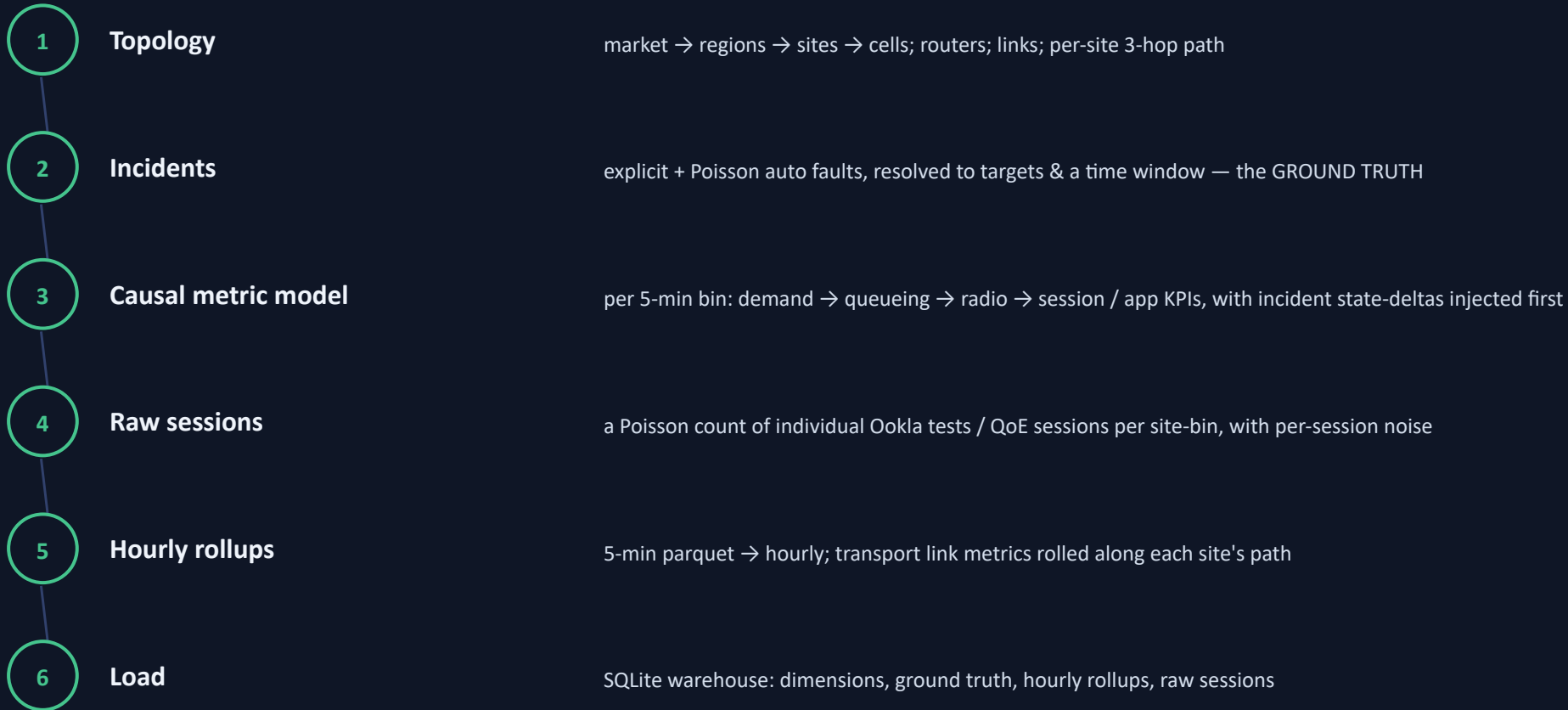
M/M/1 queueing on shared links → delay / jitter / loss

Session / App

TCP RTT, throughput, VoNR MOS (E-model), YouTube QoE

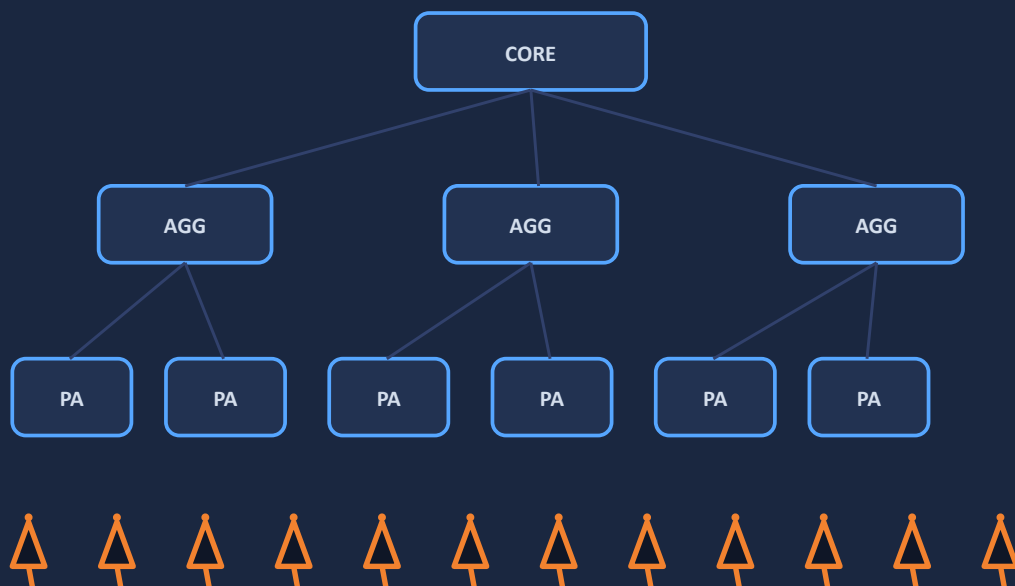
Every metric is derived from shared latent state — so the KPI correlations are physical, not bolted-on noise.

Six stages — seeded and reproducible



~1,200 sites · ~1,450 links · 5-minute granularity · 1–2 weeks · same seed → byte-identical dataset

A 3-tier transport tree with shared uplinks



site → access link → PA → PA-uplink → AGG → AGG-uplink → core

- Access link: serves 1 site (fibre or microwave)
- PA uplink: serves a whole cluster (~18 sites)
- AGG uplink: serves a whole region
- One PA-uplink fault degrades ~18 sites at once — that is the common-cause signature the analytics keys on
- No redundant paths: a fault has a clean, known blast radius to score against
- 3 sectors × 1–3 carriers per site; band plan by morphology

Statistical, ML and graph — explainable first

Baselining

robust trailing median + inter-decile range; morphology + load peer-group z-scores

Impact scoring

weighted multi-criteria composite; session- and user-impacted weighting

Attribution

rule engine $\cup$ LightGBM (gradient-boosted trees) + SHAP; isotonic-calibrated ensemble

Anomaly

STL residual + Isolation Forest + PCA reconstruction error; time + topology clustering

Correlation

Spearman, partial correlation (precision matrix), lead / lag cross-correlation

Variability

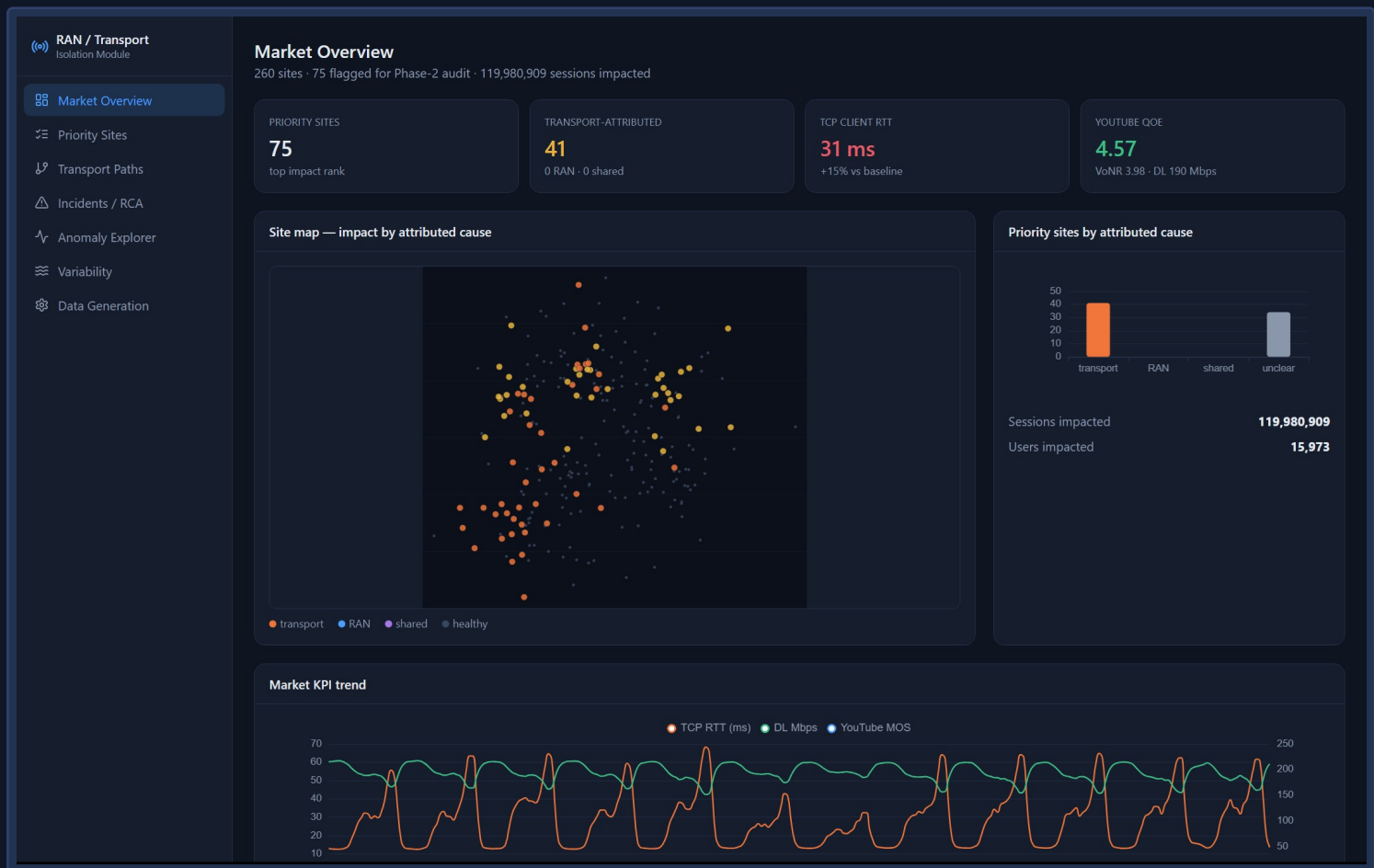
variance components, coefficient of variation, PSI drift (week over week)

Root cause

networkx transport graph + minimal hitting set (Noisy-OR fault localisation)

Explainable first: hand rules + SHAP + plain-English evidence. No LLM in the runtime.

Market overview → the priority list



Priority Sites
75 of 75 shown — ranked for Phase-2 field audit

all causes · all regions · Audit list (xlsx)

#	SITE	REGION	MORPH	IMPACT	CAUSE	CONF.	HEADLINE SEVERITY	SESSIONS	WORST WINDOW
1	S00178	R3	dense_urban	41.1	transport	0.98	🔴🔴🔴	4,575,571	(0-17 11:00)
2	S00092	R4	dense_urban	38.8	transport	0.78	🔴🔴🔴	4,318,045	(0-24 14:00)
3	S00206	R2	urban	36.0	transport	0.78	🔴🔴🔴	3,624,836	(0-17 12:00)
4	S00223	R4	urban	30.5	transport	0.88	🔴🔴🔴	4,571,492	(0-24 12:00)
5	S00004	R5	urban	26.3	transport	0.78	🔴🔴🔴	4,308,207	(0-24 15:00)
6	S00002	R4	dense_urban	25.9	transport	0.78	🔴🔴🔴	10,684,444	(0-24 15:00)
7	S00019	R1	dense_urban	21.1	transport	0.98	🔴🔴🔴	2,890,851	(0-18 16:00)
8	S00054	R5	urban	16.9	none	0.78	🔴🔴🔴	3,449,759	(0-24 16:00)
9	S00154	R4	dense_urban	15.4	transport	0.78	🔴🔴🔴	2,563,432	(0-24 16:00)
10	S00073	R6	suburban	15.3	none	0.78	🔴🔴🔴	4,141,513	(0-24 16:00)
11	S00831	R6	suburban	13.9	transport	0.78	🔴🔴🔴	2,692,609	(0-24 15:00)
12	S00187	R4	dense_urban	13.8	transport	0.78	🔴🔴🔴	3,472,463	(0-24 16:00)
13	S00040	R4	dense_urban	13.5	none	0.79	🔴🔴🔴	1,079,453	(0-24 16:00)
14	S00177	R1	suburban	13.5	transport	0.98	🔴🔴🔴	558,965	(0-27 12:00)
15	S00257	R4	dense_urban	13.1	none	0.78	🔴🔴🔴	803,067	(0-24 16:00)
16	S00105	R6	urban	13.0	none	0.79	🔴🔴🔴	919,369	(0-26 15:00)
17	S00195	R4	dense_urban	13.0	transport	0.78	🔴🔴🔴	1,724,156	(0-24 16:00)
18	S00008	R4	dense_urban	12.6	transport	0.88	🔴🔴🔴	609,914	(0-24 16:00)

Ranked top 75–100 · filter by cause / region

One-click export to the Phase-2 audit list (.xlsx)

Priority Sites — the ranked audit shortlist

Market Overview — site map coloured by attributed cause, KPI tiles vs baseline, attribution split

Per site — why this attribution



Rule evidence

plain-English, threshold-triggered — no model

SHAP

which model features drove the call

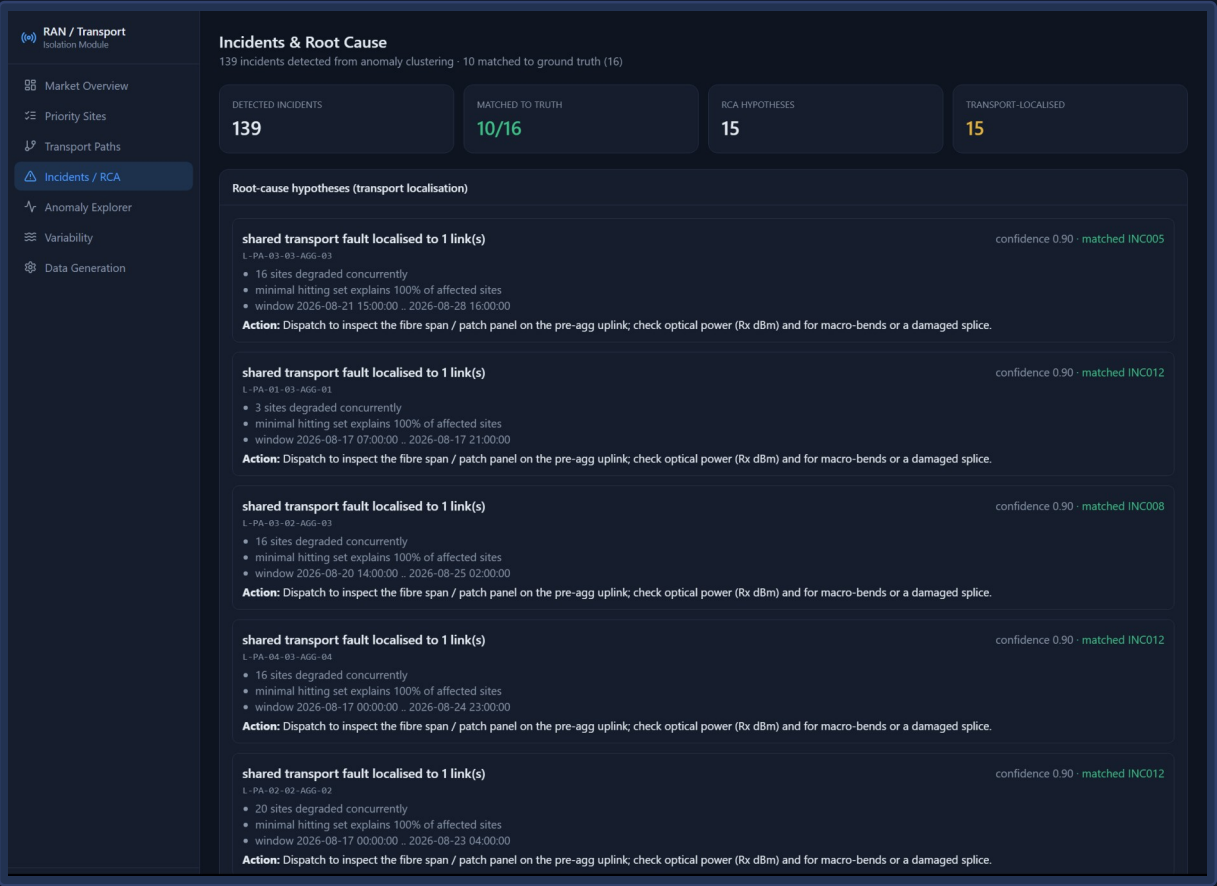
Radio vs Transport panels

side by side, with anomaly markers

Ground truth (demo)

the injected fault, so you can check it

Monday 08:10 — a monsoon fibre event



- 1 Overview**
Transport-attributed count jumps overnight; an orange cluster appears in region R2 on the map.
- 2 Priority Sites**
Filter to 'transport' — the top rows share region R2 and an 08:00–14:00 worst window. Note S00178.
- 3 Site Detail (S00178)**
Transport panel spikes (TWAMP loss +12.7 σ , SevOne queue +17.5 σ); radio flat; client-RTT up, server-RTT flat; 60 % of sibling sites also degraded. Confidence 0.98.
- 4 Transport Paths**
S00178's PA-uplink shows 6.6 % loss; almost every sibling on it is orange → the fault is the link, not 19 sites.
- 5 Incidents / RCA**
Already localised: 'shared transport fault → 1 link, explains 100 %, matched INC005'. Action: inspect the fibre span / Rx power.
- 6 Export**
Send the R2 cluster to the field team as one audit item — the link — with the recommended checks.

Standalone, controllable, explainable

Who uses it

- NOC / service assurance — daily triage
- IP transport ops — target the link
- RAN optimisation — filter transport noise
- Planning — recurring congestion → upgrades
- Field audit team — the prioritised list

Deploy & control

- docker compose up — or native (uv + Node)
- API + dashboard on :8000; SQLite only
- Windows / Linux; VS Code tasks included
- Data Generation page: presets, distributions, labelled incidents, live progress

Troubleshoot

- 'none' = degraded, cause unclear — don't dispatch
- low confidence → corroborate on Transport Paths
- no data → run generate + analytics
- reach from a laptop → bind 0.0.0.0 + open port
- OOM → fewer sites / days

github.com/manjunathsept11/ran-transport-isolation

Docs in the repo: README · DASHBOARD_GUIDE · METHODS · DATA_GENERATION · INSTALL_VM